

VL-TR: Vision Language Transformers Recommender

(Short Video Feed Deep Learning)

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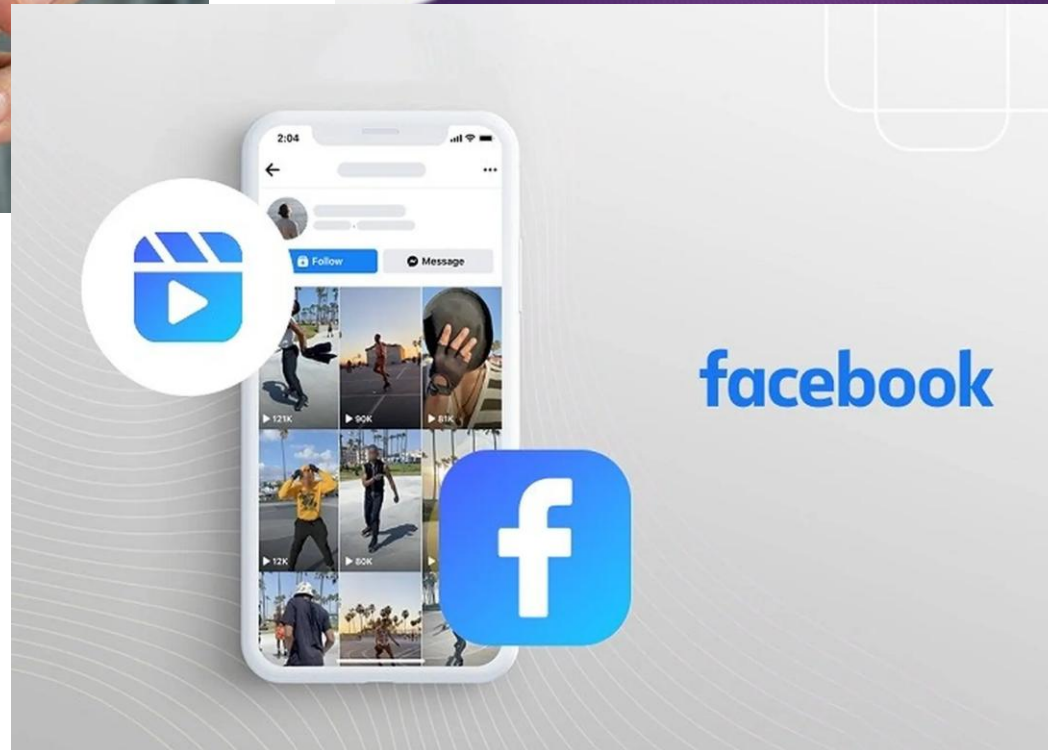
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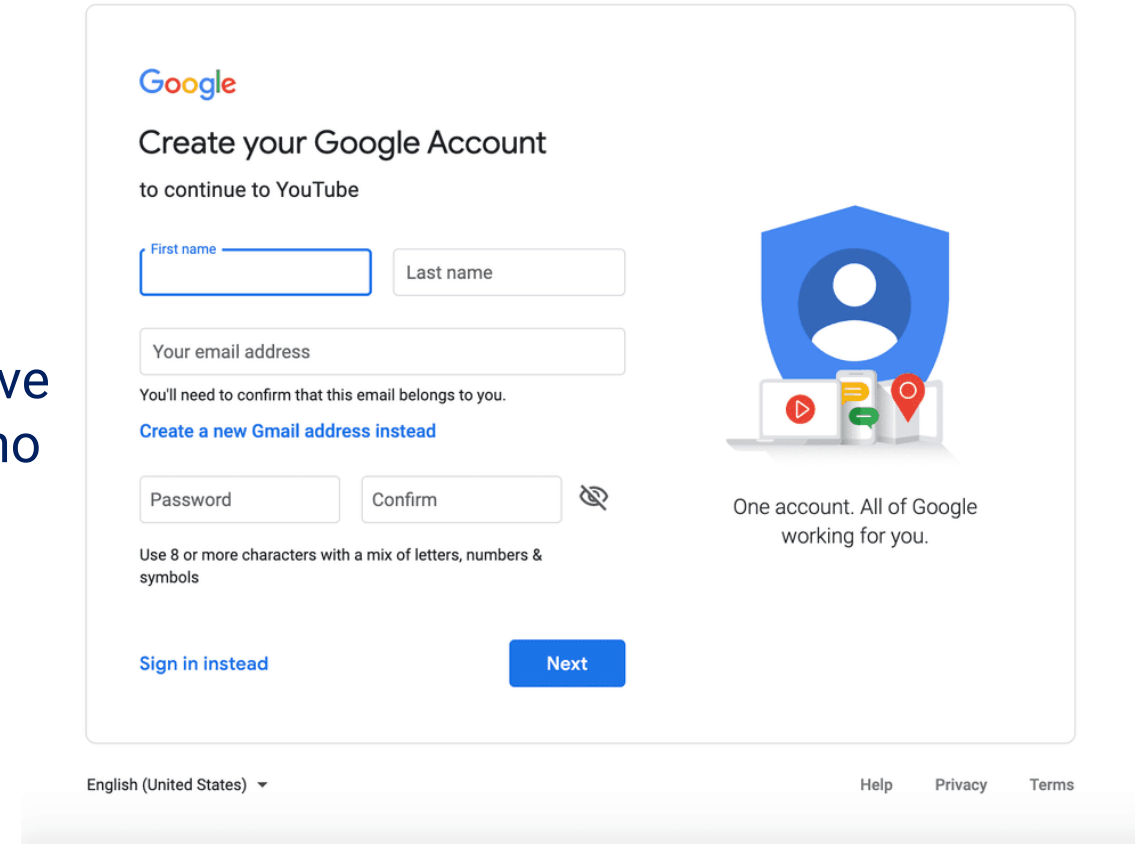


Short videos Platforms



Why Recommending Short Videos Is Hard ?

- **Trend:** The explosion of short-form video(Tiktok, Reels, Shorts, Kuaishou, Youtube Short, Facebook Reels,...)
- **Challenges:**
 - **Massive, diverse data** (Video, text, audio)
 - **Cold-start** (Core Issue): Newly uploaded videos have no interaction yet -> existing models don't know who to recommend them to
 - **Sparsity:** The user-video interaction matrix is extremely sparse



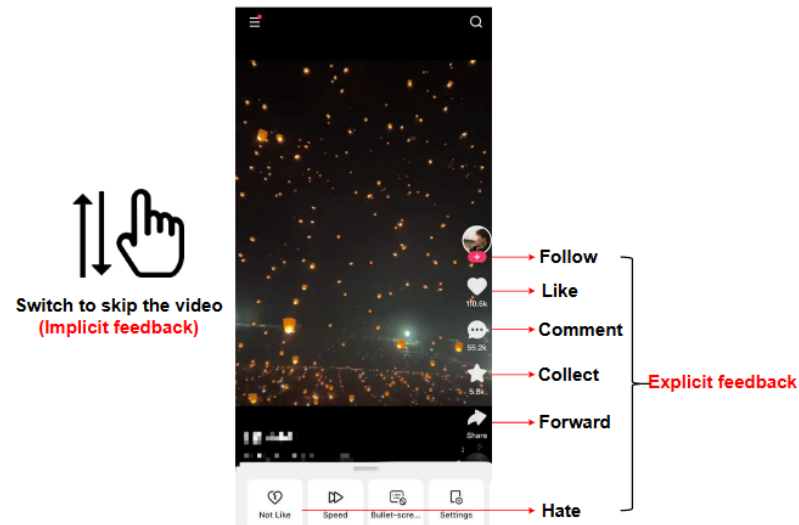
[1] Large-Scale Short-Video Platform Dataset

Scale

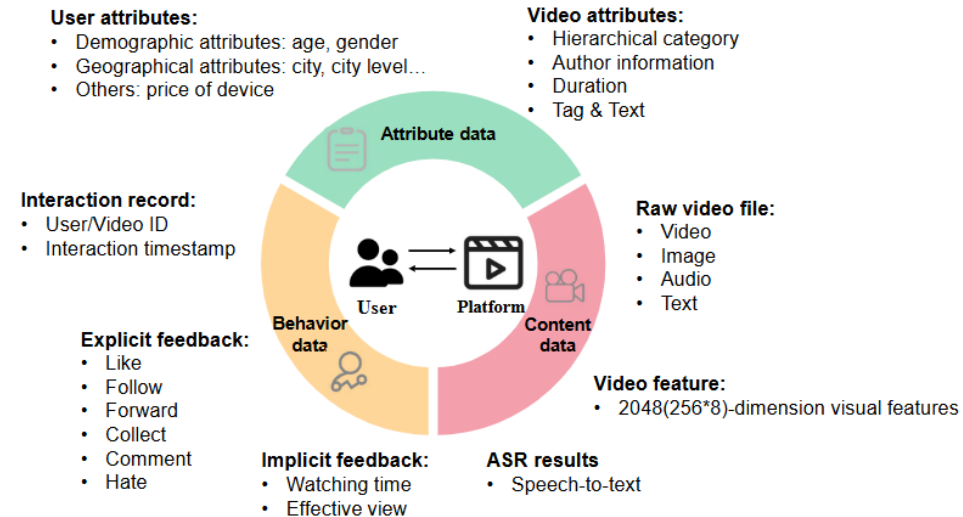
- ~ 10,000 users
- ~ 153,000 videos
- Over 1 million user-video interactions
- Highly Sparse user-video interaction Matrix

Core components

1. User-video interactions
 - Implicit Feedback: Watch time, Effective views
 - Explicit Feedback: Like, comment, share, follow, collect,...
2. Attributes
 - User attributes: Age, gender, region, device type
 - Video attributes: category, author, duration, tags, textual data
3. Multimodal Context
 - Raw video data with visual features
 - Audio Transcribed via speech-to-text



(a) User interface and behaviors on the platform.



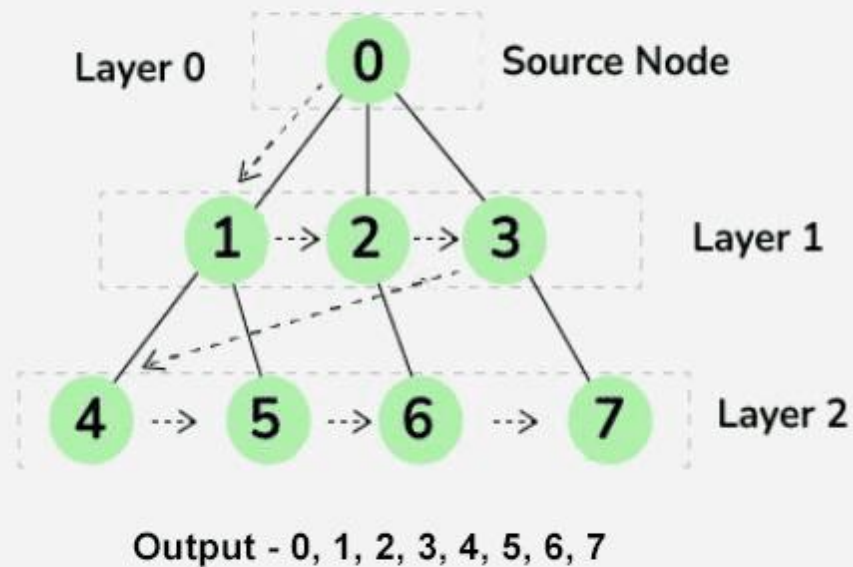
(b) Dataset Overview.

[1]Small-Scale Short-Video Platform Dataset (Custom Datasets)

- Method: Network-oriented crawling using BFS from seed users
- Purpose: Combine to Large-scale Short Video to make datasets more diversity.
- Scale: 3k videos, 10k comments

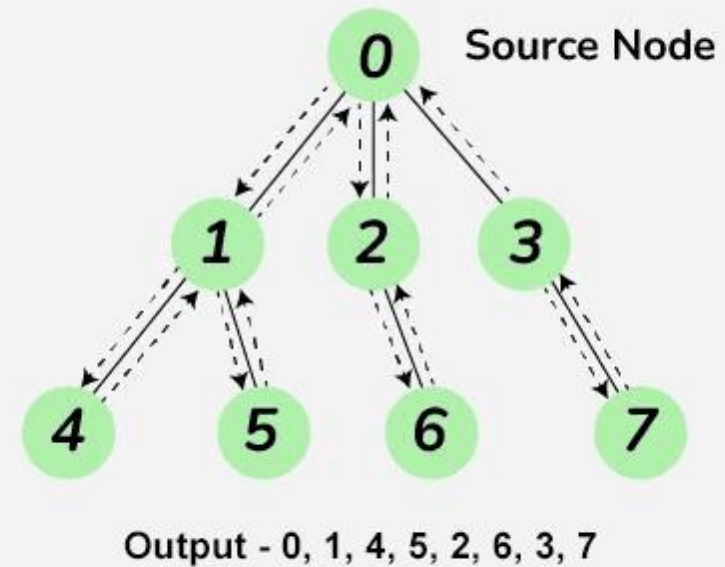


BFS



VS

DFS

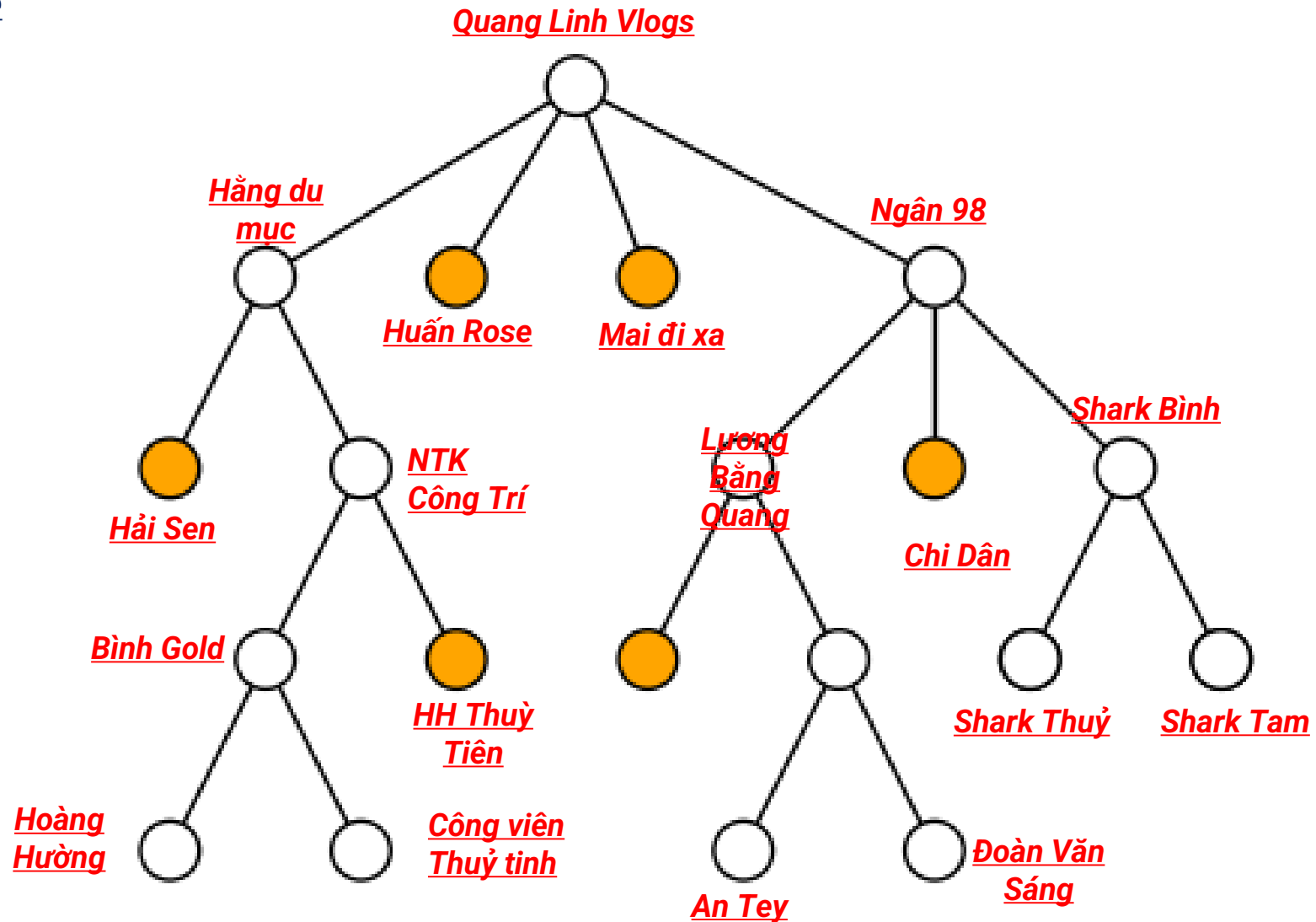


Difference Between BFS and DFS

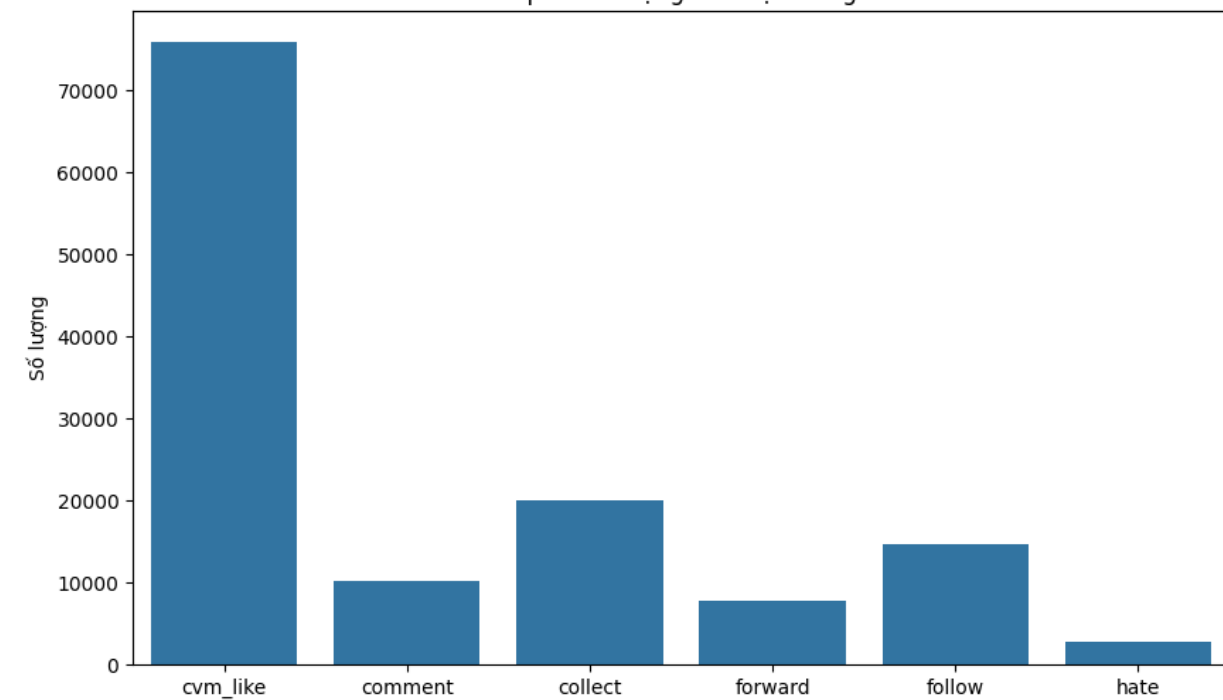


[1]Small-Scale Short-Video Platform Dataset (Custom Datasets)

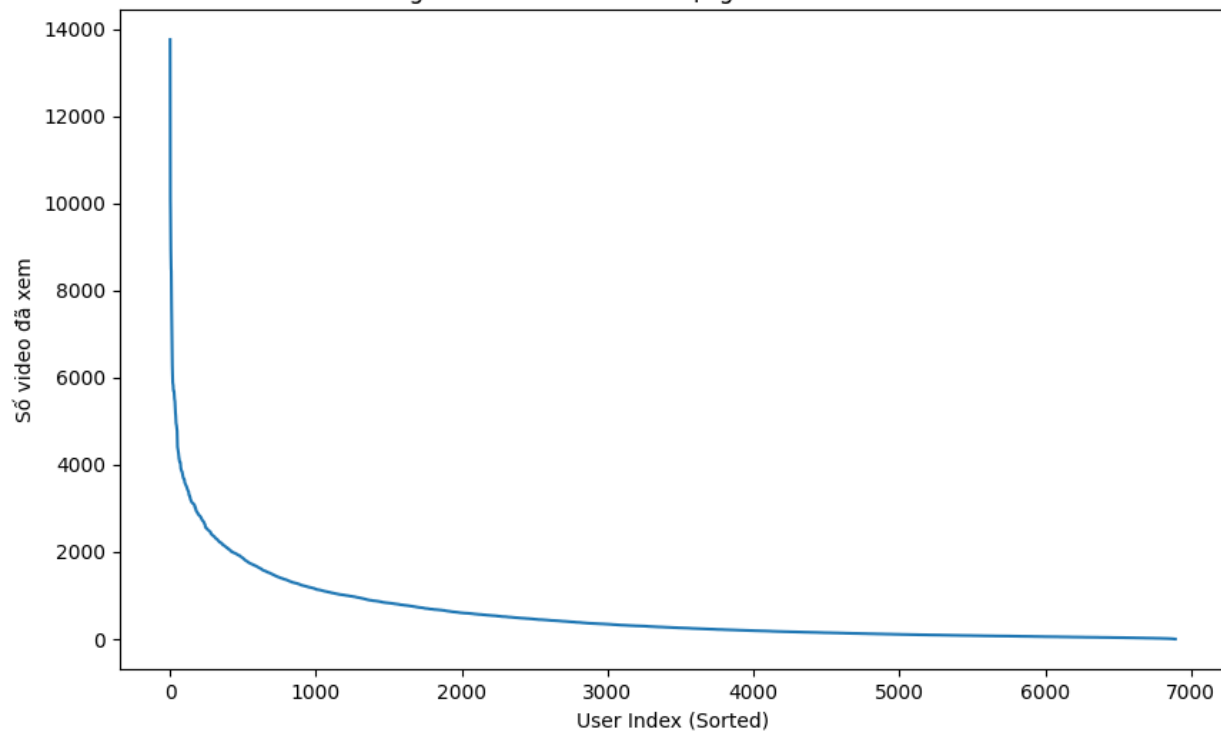
Breath First Search Tree Crawl Strategy



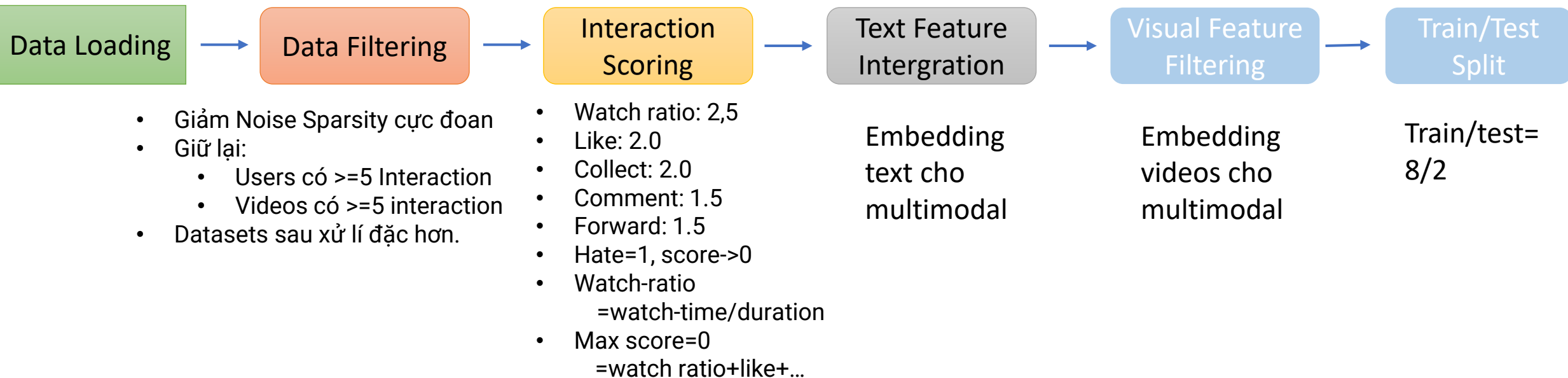
Phân phối số lượng các loại tương tác



Long-tail Distribution: Số lượng video mỗi User xem



Data Preprocessing



Matrix Factorization (Singular Value Decomposition) – ML Baseline

Type: Collaborative Filtering

Pros:

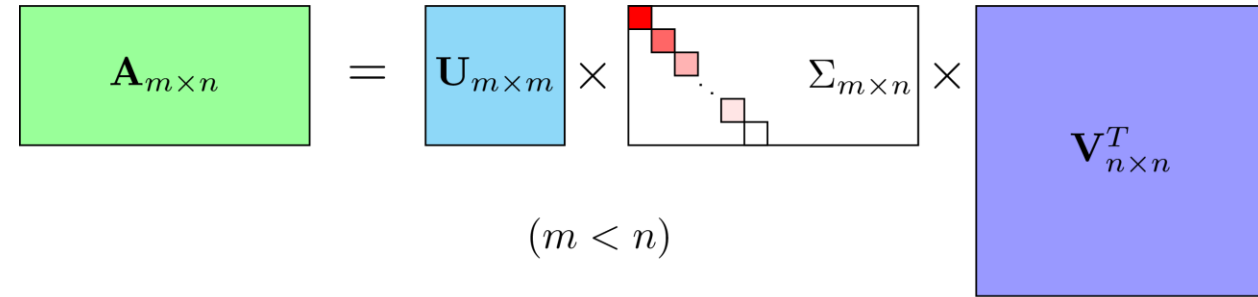
- Good at discovering hidden user preferences.
- Fast and scalable.
- Bias terms reduce noise from extreme users or very popular items.

Cons:

- Only captures simple (linear) interactions.
- Content-blind (uses only IDs, no understanding of video content).
- Suffers badly from the cold-start problem.

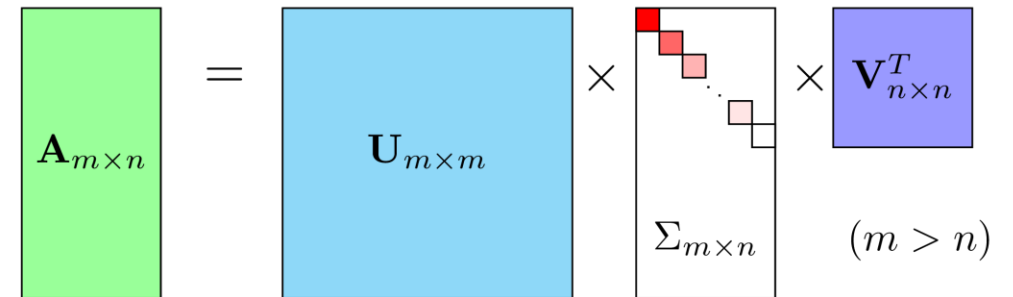
$$\mathbf{A}_{m \times n} = \mathbf{U}_{m \times m} \times \Sigma_{m \times n} \times \mathbf{V}_{n \times n}^T$$

$(m < n)$



$$\mathbf{A}_{m \times n} = \mathbf{U}_{m \times m} \times \Sigma_{m \times n} \times \mathbf{V}_{n \times n}^T$$

$(m > n)$



Gradient Boosting (LightGBM – Learning to Rank) – ML Baseline

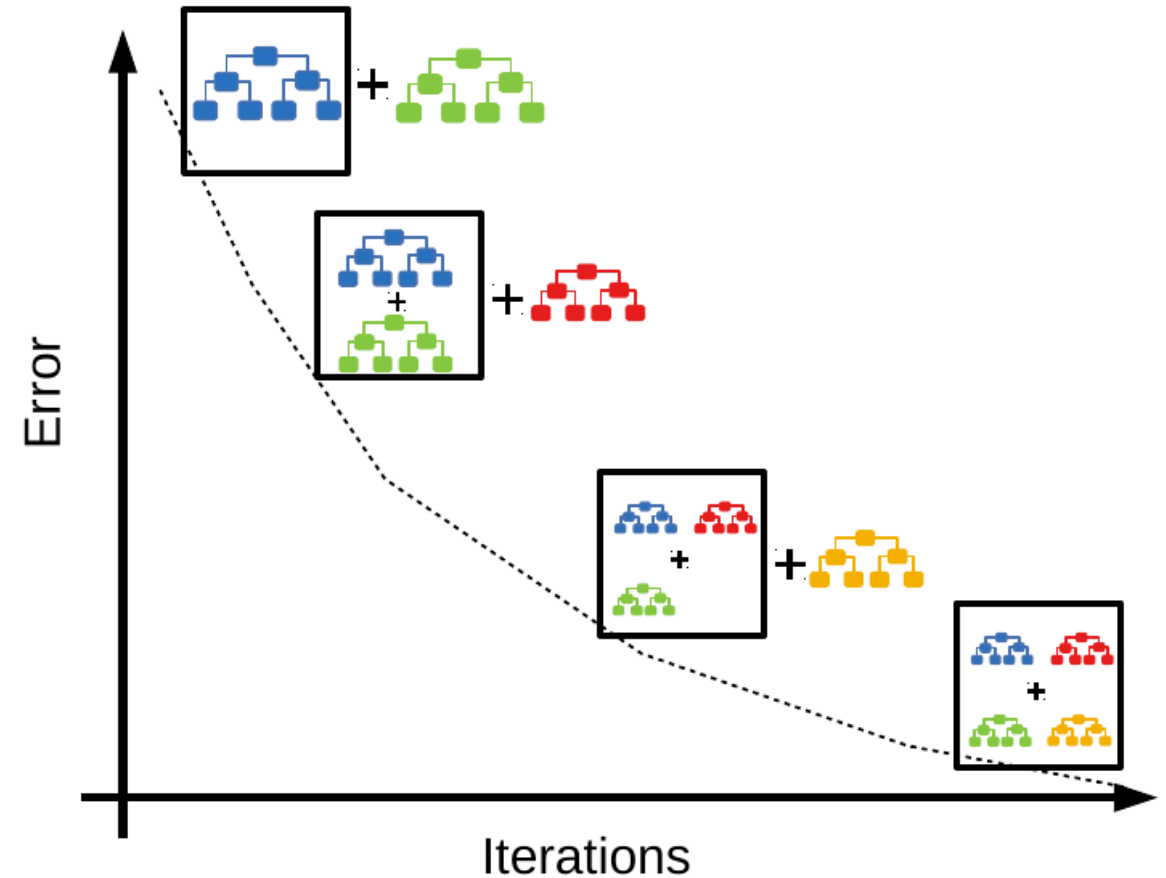
Mechanism: Uses sequential decision trees with LambdaRank, relying on manual feature engineering.

Pros:

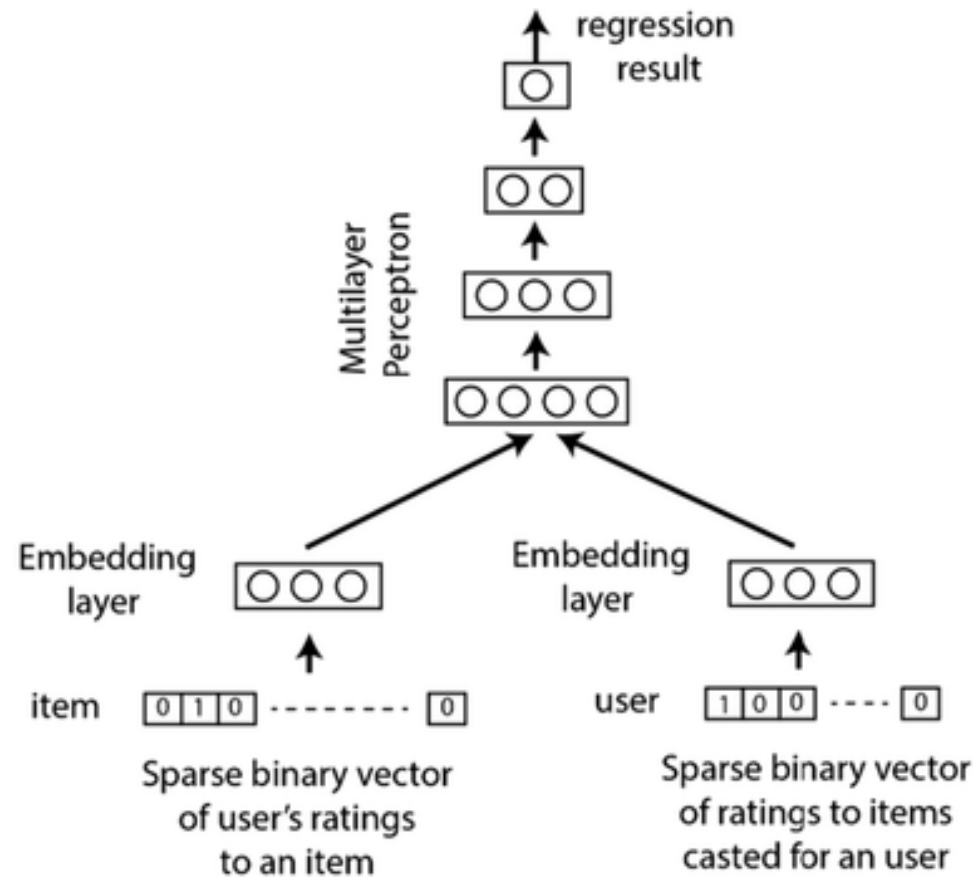
- Directly optimizes ranking metrics (NDCG/MAP).
- Handles both numerical and categorical data.
- Interpretable via feature importance.

Cons:

- Performance depends heavily on feature engineering.
- Cannot learn features from raw data (no representation learning).



Neural Collaborative Filtering (NCF) – DL Baseline



Neural Collaborative filtering (NCF)

Mechanism: Replaces the dot product in SVD with a multi-layer neural network (MLP) to learn the interaction function from two embedding vectors (User ID & Item ID).

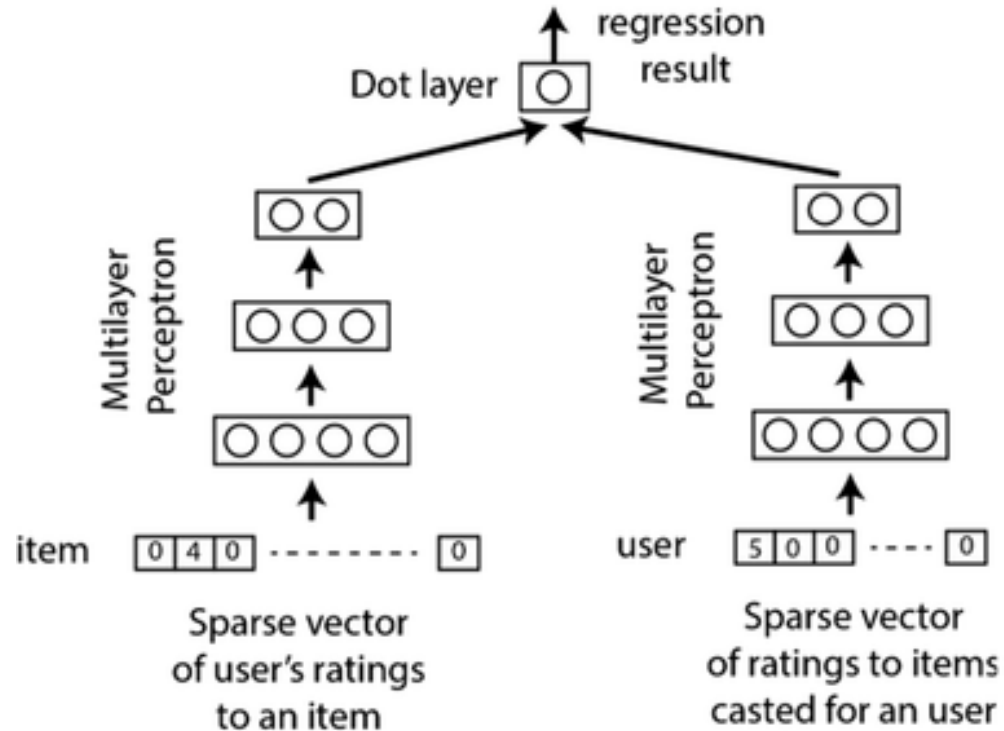
Advantages:

- **Non-linearity:** Thanks to activation functions (ReLU, Sigmoid), NCF can learn more complex and subtle user–item interactions that SVD misses.
- **Generalization ability:** A flexible framework that is easy to scale in depth and width.

Disadvantages:

- **Still ID-based:** It relies only on IDs, so the cold-start problem remains just like in SVD.
- **Harder to train:** MLP networks can be harder to converge than SVD without careful tuning.

Deep Matrix Factorization(DeepMF) – DL Baseline



Deep Matrix Factorization (DeepMF)

Mechanism: Combines two branches: a linear branch (like SVD) and a non-linear branch (like NCF).

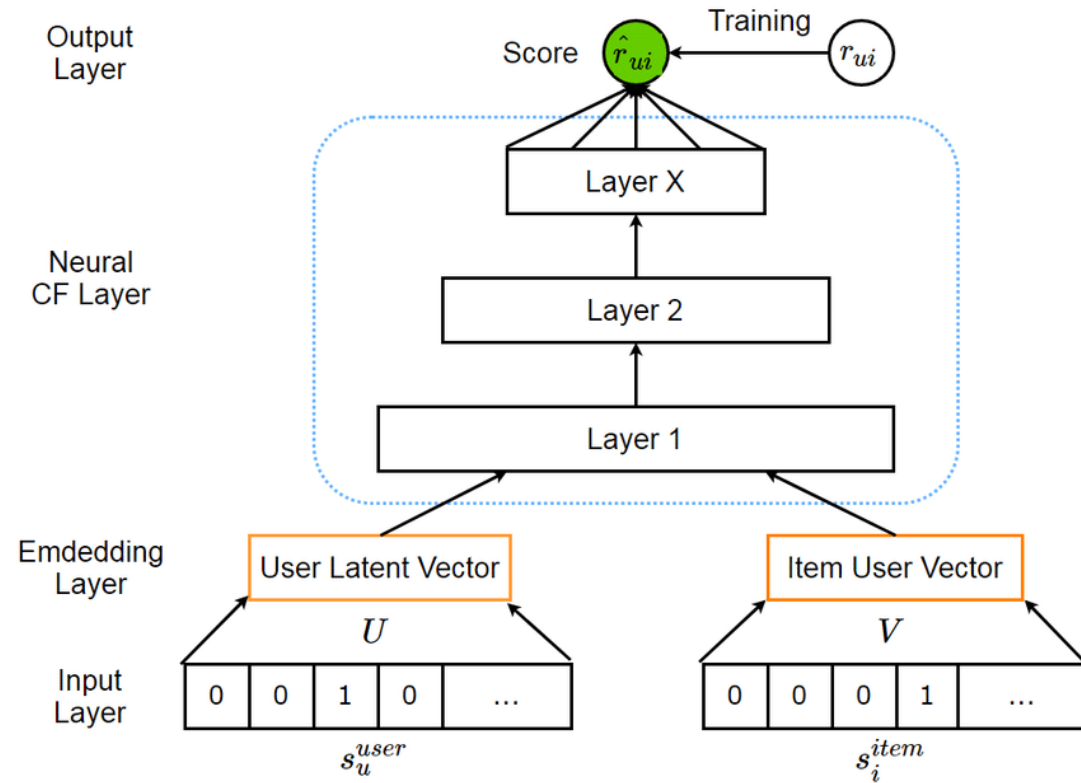
Advantages:

- Leverages the best of both worlds: captures simple interactions (linear) while also learning complex ones.
- Usually more stable than pure NCF.

Disadvantages:

- Doubles the number of parameters (requires two separate embedding sets), making it more resource-intensive.
- Still does not solve the root cause of the cold-start problem.

Content-Enhanced NCF (ContentNCF)



Mechanism: Concatenates a text vector (text embedding from the title/description) to the input of NCF.

Advantages:

- **Semantic understanding:** Starts to capture the “topic” of a video through its title.
- **Mitigates cold-start:** If a new video has a title similar to existing ones, the model can make some reasonable recommendations.

Disadvantages:

- **Visual gap:** For short videos (e.g., TikTok), visuals matter more than text. ContentNCF completely ignores the visual aspect.
- **Naive concatenation:** Simply appending text vectors to ID embeddings does not help the model learn deeper correlations (e.g., when the text says A but the video shows B).

Issue

Why do we propose the VL-TR model instead of using the above approaches?

1. Thoroughly addressing the cold-start problem

Problem: Traditional models (SVD, NCF, DeepMF) struggle with new videos because they have no interaction history yet.

VL-TR solution: Instead of relying on IDs, VL-TR uses a Vision Encoder for visuals and a Text Encoder for titles as soon as a video is uploaded.

2. Capturing multimodal understanding

Problem: Short videos combine both vision and language, so using only text (like in ContentNCF) is insufficient.

VL-TR solution: A Transformer processes both modalities together, and Cross-Modal Attention helps link images with text (e.g., “Cute cat” matches a cat image).

3. Superior prediction precision

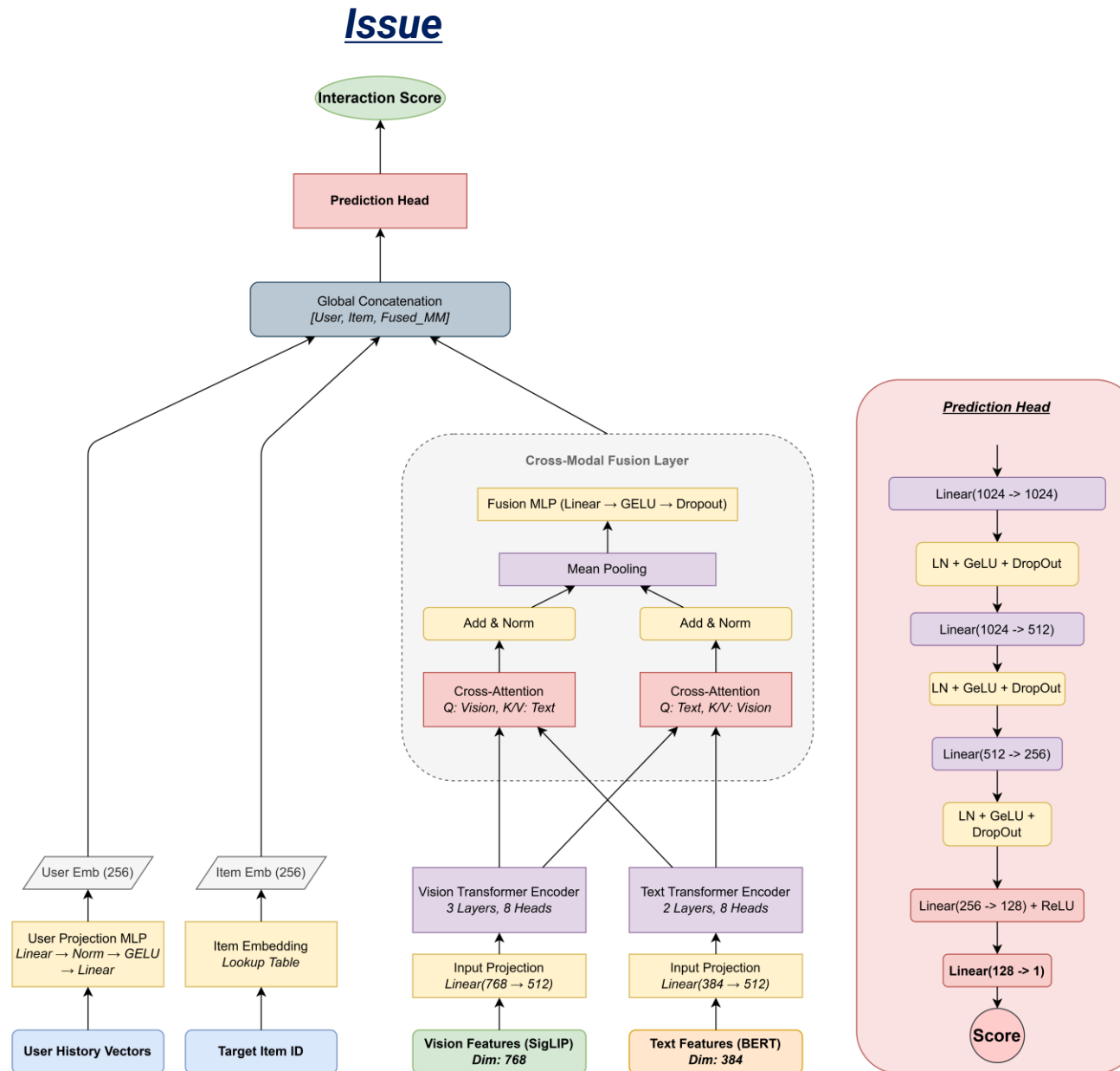
Problem: Traditional models rank by popularity but are weak at predicting exact preference levels (high RMSE).

VL-TR solution: With deeper content understanding, VL-TR achieves much lower error ($\text{RMSE} \approx 0.47$) and can tell casual watching from full engagement.

4. Modern Transformer-based architecture

Instead of static features like LightGBM, VL-TR uses Transformer Encoders that support fine-tuning and richer representations, making them stronger than MLP-based models.

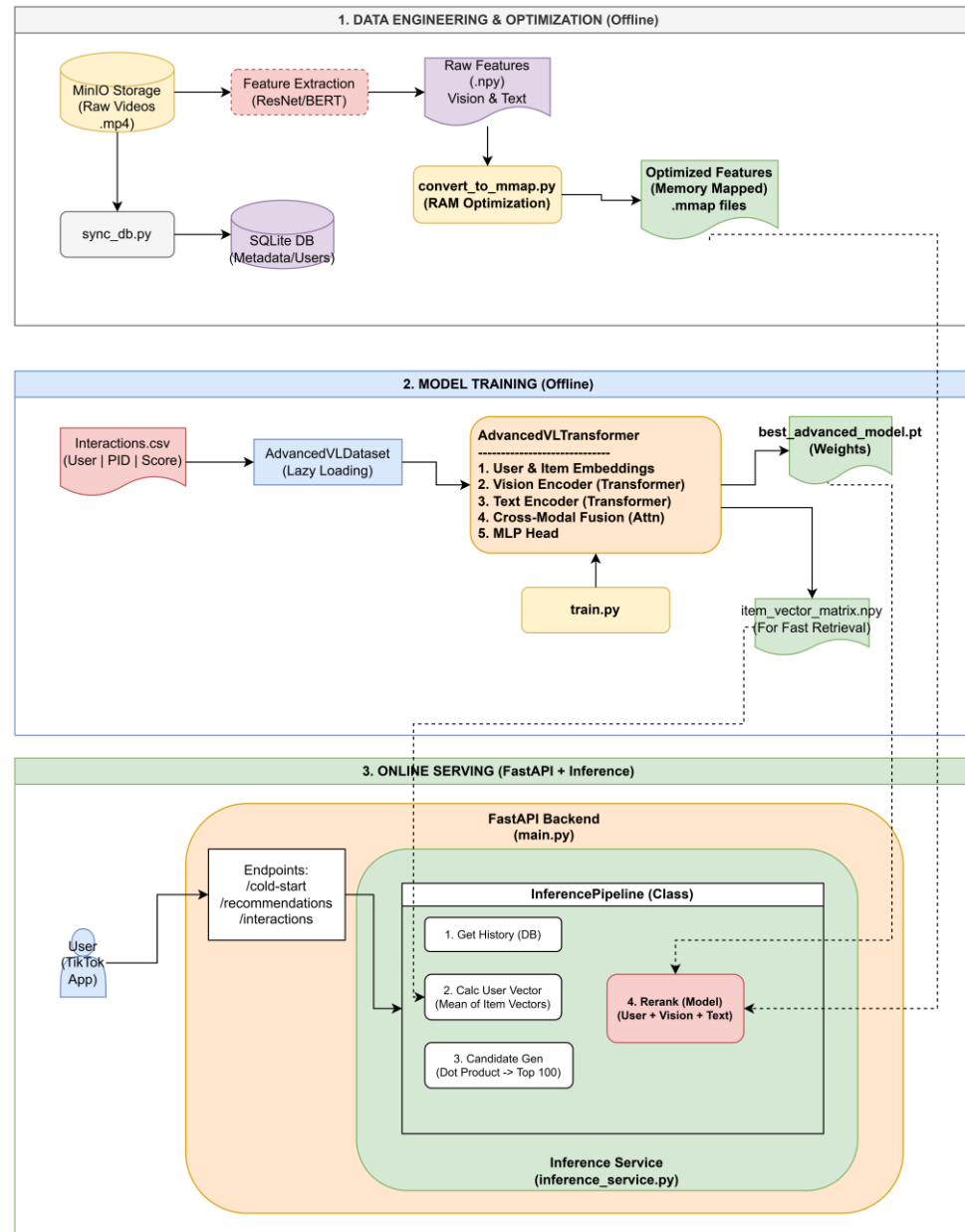
Proposed Method



Criteria	Machine Learning Baselines (SVD, LightGBM)	Deep Learning Baselines (NCF, DeepMF)	Proposed Method (VL-TR)
1. Input Representation	Hand-crafted Features: Statistical aggregates (mean/std), TF-IDF vectors, and interaction counts. Relies on manual feature engineering.	ID Embeddings: Sparse User IDs and Item IDs are mapped to dense learnable vectors via Lookup Tables. Static text features may be concatenated.	Raw Multimodal Content: Processes raw visual frames (via ViT) and textual tokens (via BERT) directly. User preferences are learned relative to content semantics.
2. Optimization Algorithm	SGD / Gradient Boosting: SVD uses Stochastic Gradient Descent. LightGBM uses Histogram-based Gradient Boosting.	Adam: Standard adaptive moment estimation with fixed momentum parameters.	AdamW + Scheduler: Uses Adam with Weight Decay ($1e-4$) to prevent overfitting on complex Transformer layers. Integrates ReduceLROnPlateau for dynamic learning rate adjustment.
3. Loss Function	LambdaRank / Squared Error: LightGBM optimizes ranking metrics (NDCG) directly via gradients. SVD minimizes reconstruction error.	MSE / LogLoss: Minimizes the point-wise error between predicted scores and ground truth ratings.	End-to-End MSE: Optimizes the entire pipeline (Encoders + Fusion + MLP) simultaneously based on interaction intensity errors.
4. Data Management	In-Memory: Requires loading the entire dataset and feature matrix into RAM before training. Limited scalability.	Standard Batching: Loads batches of IDs into GPU memory. Relatively lightweight due to small embedding sizes.	Memory-Mapped (Zero-Copy): Uses <code>numpy.memmap</code> to stream high-dimensional visual tensors from SSD to RAM on-demand. Enabling training on datasets larger than physical memory.
5. Negative Sampling	Static Ratio (1:2): For every positive interaction, 2 negative samples are generated once or sampled simply.	Random Ratio (1:4): Standard 1 positive to 4 negatives strategy to balance the dataset.	Dynamic Batching (1:4): Negatives are sampled dynamically during training to ensure the model sees a diverse range of "non-relevant" content.
6. Learning Mechanism	Memorization: Excellent at memorizing existing user-item pairs and popularity biases.	Non-linear Interaction: Learns complex interaction functions via MLP but remains ID-dependent.	Representation Learning: Learns to "understand" video content via Attention mechanisms. Fine-tunes feature extractors end-to-end.
7. Cold-Start Handling	None: Cannot handle new items (requires retraining to generate new latent factors).	Limited: Fails for new IDs (assigns random embedding). ContentNCF helps slightly via static text.	Full Support (Zero-Shot): Can generate recommendations for brand new videos immediately based on Visual/Textual content analysis.

Method	Rec@10	Rec@20	NDCG@10	NDCG@20	Prec	MAP	HitRate	RMSE ↓	MAE ↓
<i>Machine Learning Baselines (Feature Engineering)</i>									
SVD-Dominant	<u>0.9655</u>	0.9957	<u>0.7763</u>	<u>0.7857</u>	<u>0.1507</u>	<u>0.7057</u>	0.9812	1.0419	<u>0.8244</u>
LGBM Ranker	0.9696	0.9970	0.7871	0.7956	0.1516	0.7190	0.9839	<u>1.0393</u>	0.8316
Confidence-Hybrid	0.9652	<u>0.9961</u>	0.7606	0.7705	0.1502	0.6853	<u>0.9826</u>	1.0699	0.8891
<i>Deep Learning Baselines (Latent Factor Models)</i>									
NeuralCF (NCF)	0.9622	0.9930	0.7346	0.7443	0.1492	0.6539	0.9826	1.1779	0.9415
DeepMF	0.9485	0.9940	0.7312	0.7455	0.1468	0.6559	0.9732	1.2019	0.9676
ContentNCF	0.9620	0.9951	0.7536	0.7641	0.1494	0.6775	0.9812	1.1638	0.9330
<i>Proposed Multimodal Method</i>									
VL-Transformer	0.6503 [†]	0.8539 [†]	0.3647 [†]	0.4163 [†]	0.0650	0.2783	0.6503	0.4703	0.2506

[†] *Note on Evaluation Protocols:* The baseline models were evaluated on a standard test set primarily consisting of warm-start items (items with history). In contrast, the VL-Transformer was subjected to the rigorous "**1 Positive vs. 99 Negatives**" protocol, a significantly harder "needle-in-a-haystack" task designed to test robustness against noise, explaining the divergence in raw ranking scores.



- [SVD]** Koren, Y., Bell, R., & Volinsky, C. (2009). *Matrix factorization techniques for recommender systems*. IEEE Computer.
- [LightGBM]** Ke, G., et al. (2017). *LightGBM: A highly efficient gradient boosting decision tree*. NIPS.
- [NCF]** He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T. S. (2017). *Neural collaborative filtering*. WWW '17.
- [DeepMF]** Xue, H. J., et al. (2017). *Deep Matrix Factorization Models for Recommender Systems*. IJCAI.
- [Transformer]** Vaswani, A., et al. (2017). *Attention is all you need*. NIPS.
- [ViT]** Dosovitskiy, A., et al. (2021). *An image is worth 16x16 words: Transformers for image recognition at scale*. ICLR.
- [S-BERT]** Reimers, N., & Gurevych, I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*. EMNLP.

Thank You !

Question?